

Experiment No : 05

Experiment Name : Determination of the radius of curvature of a plano convex lens by Newton's Ring method

Theory:

When monochromatic light of wavelength λ is incident on a plano convex lens placed on a plane glass plate, a thin air film is formed between them. Light reflected from the lower surface of the lens and the upper surface of the glass plate interferes, producing alternate bright and dark concentric rings known as Newton's rings. One of the reflected rays undergoes a phase change of π , resulting in a dark ring at the center due to zero air film thickness.

Let, D_n and D_{n+p} be the diameters of the n^{th} and $(n+p)^{\text{th}}$ bright or dark rings for normal incidence.

Diameter of n^{th} dark ring, $D_n = \sqrt{4n\lambda R}$ (i)

Diameter of $(n+p)^{\text{th}}$ dark ring, $D_{n+p} = \sqrt{4(n+p)\lambda R}$ (ii)

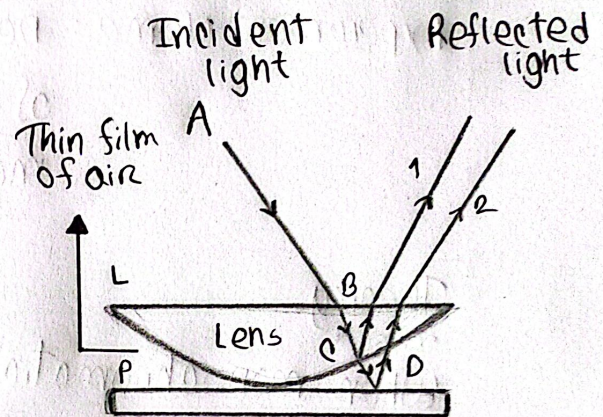
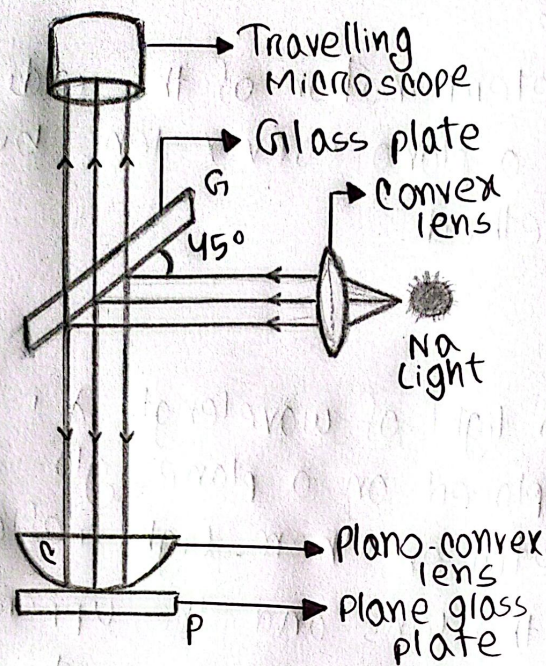
From (i) and (ii),

$$R = \frac{D_{n+p}^2 - D_n^2}{4p\lambda}$$

where,

R = Radius of curvature of plano convex lens

Thus, R can be determined by measuring the diameters of Newton's rings.



1. Reflected ray from lower surface of lens
2. Reflected ray from upper surface of glass plate

Fig 1: Experimental setup to find Newton's Ring

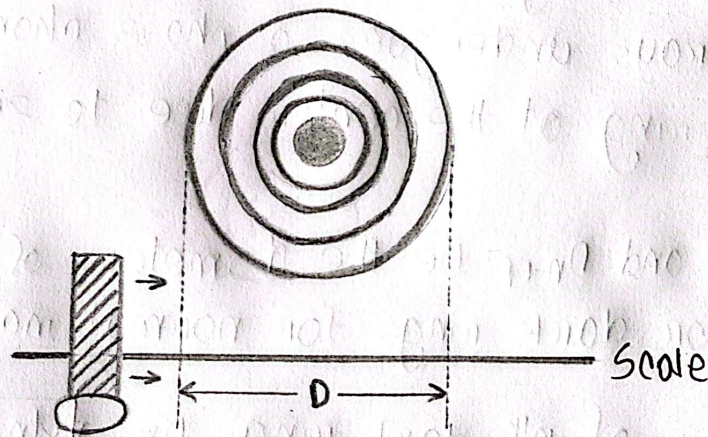


Fig 2: Observed Newton's Rings

Required Apparatus:

- (i) Plano convex lens
- (ii) Glass Plate
- (iii) Microscope
- (iv) Wooden box
- (v) Na flame arrangement

Calculations:

Experimental Data:

Table 1: Readings for diameters of the rings

Least count (LC) = cm

Ring no.	Readings of the microscope								Diameter, $D = (L - R)$ (cm)	D^2 (cm ²)
	LEFT SIDE				RIGHT SIDE					
	MSR, S (cm)	CSO, D (div)	CSR, C = (D × LC) (cm)	TR, L = (S + C) (cm)	MSR, S (cm)	CSO, D (div)	CSR, C = (D × LC) (cm)	TR, L = (S + C) (cm)		
1										
2										
3										
4										
5										
6										
7										
8										

Results:

R =

..... (from calculation)

R =

..... (from graph)

Precautions:

- Backlash error was avoided
- Diameter was measured very carefully
- The cross wire of travelling microscope was set mid-way between the outer and inner edges of a ring to avoid error

Name: Roll No:

